

I3DION Spatial

Feature Specification Document (FSD) v1.0

Product Name

I3DION Spatial

Product Category

Industrial AR Catalog & Sales Visualization Platform

Document Version

1.0

Feature 01 – Product Management

Purpose

The Product Management Module serves as the foundation of the I3DION Spatial platform. It allows companies to create, organize, maintain, and publish digital representations of their products.

Every experience within the platform originates from a product record.

Business Value

This module enables companies to maintain a centralized product repository that can be reused across catalogs, QR experiences, AR visualizations, and marketing campaigns.

Functional Requirements

Create Product

Users must be able to create a new product.

Required Fields:

- Product Name

- Product Category
- Product Description
- Product Specifications
- Product Thumbnail

Optional Fields:

- Product Videos
 - Product Documents
 - Product Tags
-

Upload Assets

Supported File Types:

Images:

- JPG
- PNG
- WEBP

Videos:

- MP4

3D Models:

- GLB

Documents:

- PDF
-

Product Status

Available States:

- Draft
 - Published
 - Archived
-

Feature 02 – Catalog Builder

Purpose

The Catalog Builder enables companies to create interactive digital catalogs using products stored in the platform.

Business Value

Traditional PDF catalogs are static.

I3DION Spatial transforms catalogs into interactive experiences that connect physical products with digital visualization.

Functional Requirements

Create Catalog

Users can:

- Create Catalog
 - Edit Catalog
 - Delete Catalog
-

Catalog Information

Fields:

- Catalog Name
 - Description
 - Category
 - Cover Image
-

Product Assignment

Users can:

- Add Products
- Remove Products
- Reorder Products

Catalog Publishing

Output:

- Public URL
 - QR Code
 - PDF Export
-

Feature 03 – QR Engine

Purpose

The QR Engine acts as the bridge between physical and digital experiences.

Business Value

Sales teams can distribute catalogs through:

- Printed Brochures
 - Product Cards
 - Sales Documents
 - Product Stickers
-

Functional Requirements

Product QR

Generate a unique QR code for each product.

Catalog QR

Generate a unique QR code for each catalog.

QR Actions

When scanned:

QR Scan

↓

Landing Page

↓

Product Experience

QR Export

Formats:

- PNG
- SVG

Feature 04 – AR Experience

Purpose

The AR Experience allows users to visualize products in their real-world environment.

Business Value

Customers can understand:

- Product Size
- Product Placement
- Product Appearance

before making purchasing decisions.

Functional Requirements

AR Launch

User clicks:

"View in AR"

Placement

Users can:

- Place Product
 - Move Product
 - Rotate Product
 - Scale Product
-

Environment Detection

The system should:

- Detect Horizontal Surfaces
 - Detect Placement Areas
-

Device Support

Supported:

- Android ARCore
 - iOS Quick Look
 - WebXR
-

Feature 05 – Lock / Unlock System

Purpose

Allow products to remain fixed in a specific real-world location.

Business Value

Customers can walk around products and inspect them naturally.

Workflow

Place Product

↓

Lock Position
↓
Move Around Product
↓
Inspect Product
↓
Unlock Position

Functional Requirements

Lock Product

Freeze product position.

Unlock Product

Allow repositioning.

Reset Product

Return product to initial placement state.

Feature 06 – Animation Engine

Purpose

Allow companies to demonstrate product behavior through animations.

Business Value

Improves understanding of:

- Functionality
 - Operation
 - Maintenance
-

Supported Animation Types

Operational

Examples:

- Motor Rotation
 - Fan Rotation
 - Conveyor Movement
-

Maintenance

Examples:

- Door Open
 - Door Close
-

Demonstration

Examples:

- Product Workflow
 - Internal Process
-

Controls

- Play
 - Pause
 - Restart
-

Feature 07 – Hotspot Engine

Purpose

Provide contextual information directly within the product experience.

Business Value

Customers receive information without leaving the experience.

Functional Requirements

Create Hotspot

Fields:

- Title
- Description
- Position

Interaction

User clicks hotspot.

Information panel opens.

Example

Motor

15 HP Premium Motor

Tank

500 Liter Capacity

Feature 08 – CTA Engine

Purpose

Enable companies to create actions that support sales conversion.

Business Value

Convert engagement into business opportunities.

Supported Actions

Contact Sales

Launch phone call.

WhatsApp

Open WhatsApp conversation.

Request Quote

Submit inquiry form.

Download Brochure

Download PDF.

Visit Website

Open company website.

Custom CTA Support

Companies can configure custom buttons.

Feature 09 – Lead Management

Purpose

Capture and manage customer interest.

Business Value

Generate measurable sales opportunities from catalog engagement.

Lead Capture Sources

- Request Quote
 - Contact Form
 - Product Inquiry
-

Lead Information

Fields:

- Name
 - Email
 - Phone
 - Company Name
 - Interested Product
 - Message
-

Lead Status

Available States:

- New
 - Contacted
 - Qualified
 - Proposal Sent
 - Closed
-

Notifications

The system should notify company administrators whenever a new lead is generated.

Feature Dependencies

Product Management ↓ Catalog Builder ↓ QR Engine ↓ Product Experience ↓ AR Experience ↓ Lead Generation

MVP Priority

Phase 1

- Product Management
 - Catalog Builder
 - QR Engine
-

Phase 2

- AR Experience
 - Lock System
-

Phase 3

- Animation Engine
 - Hotspots
-

Phase 4

- CTA Engine
 - Lead Management
-

Expected Outcome

The combination of Product Management, Catalog Builder, QR Experiences, AR Visualization, Product Locking, Interactive Animations, Hotspots, and Lead Generation establishes the foundation of I3DION Spatial as a complete Industrial AR Catalog & Sales Visualization Platform rather than a standalone AR viewer.